

INSTRUCTIONS: Attempt ALL QUESTIONS

Question 1

{10 Marks}

Booker University library maintains an inventory of books. The list includes the details: **Author**, **price**, **title**, **book_number** and **number of copies** of each book. Whenever new books are purchased the librarian adds the book details into the database. The chief librarian occasionally requests for a list of all the books in the database.

Required

Construct a simple class called book with suitable data members and member functions to:-

- (i) Insert a new book record into the database
- (ii) Display a list of all books in the database
- (iii) Write a main function to test the program

Create a database to store the relevant records

Question 2

{10 Marks}

DT Dobie (K) Ltd sells a range of vehicles, including Mercedes and Nissan in the East African region. Vehicle sale details include the Make (e.g. Nissan), Model (e.g. Sunny), Engine Number and Sale Price.

- (i) Create a class vehicle that captures the above data using a function called set_vehicle()
- (ii) The company makes a profit of 15% from the sale price for every vehicle. Create another function called get_profit() which is still a member of the class vehicle to calculate and return the profit.
- (iii) Implement (i) and (ii) using an object in the main() function to capture the vehicle details and display the profit.
- (iv) Create a database to store the relevant records